



VOLTEDGE

Project Documentation — Executive Summary

Consumer-electronics analytics — condensed reference

Scope 10 sections condensed from the full documentation set (~100 pages → ~15)

Compiled September 2026

Company VoltEdge Electronics (fictional) — synthetic, seeded data

Full set VoltEdge-Electronics-Documentation.pdf

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01 Business Context & KPI Framework

VoltEdge Electronics — fictional pure-play online consumer-electronics retailer (direct-to-consumer, no physical stores). HQ Austin TX; reporting currency **USD**. Analytics extract window **2024-07-01 → 2026-06-30 (24 months)** for a clean year-over-year read: CY = 2025-07→2026-06, PY = 2024-07→2025-06.

Markets — scaled 1 → 4 inside the window

Market	Currency	Live since	In extract
United States	USD	always-on	full 24 months
United Kingdom	GBP	2025-01-01	18 months
Germany	EUR	2025-01-01	18 months
Brazil	BRL	2025-07-01	12 months

Revenue *steps up* as markets go live; blended metrics (CAC, AOV, margin, delivery time) shift with the market mix. Every growth metric is therefore shown three ways: **Total**, **Comparable Base** (US only — live the whole CY+PY span, i.e. like-for-like) and **Expansion Contribution** (Total – Comparable Base).

Catalog & economics

~300 active SKUs (Category → Subcategory → Brand → Product). It is a **thin-margin** business: smartphones / laptops / consoles ≈ 8–16 % gross margin; audio / accessories 22–55 %. The recurring executive question — *are we growing revenue or growing profit?* — turns on **category mix**, so price–volume–mix analysis is first-class.

Revenue = product sales (bulk) + shipping fees + extended-warranty attach (modelled as a separate `line_type = 'warranty'` line). **Costs** = COGS (moving-average), discounts, returns (8–14 %), marketing, carrier shipping cost, payment processing (+ financing of interest-free BR installments), inventory purchases.

Financial definitions

Term	Definition
Gross Revenue (GMV)	Σ (list price × qty), before discounts, excl. tax & shipping
Net Revenue	North Star — Gross Revenue – Discounts – Returns, in USD
COGS	Σ (moving-avg unit cost × net qty sold)
Gross Profit	Net Revenue – COGS (+ returns COGS recovered)
Contribution Margin	Gross Profit – Marketing – Shipping Cost – Payment Cost (operating-contribution proxy)

CFO threads modelled in depth

- Payments** — MDR by scheme, fixed fees, settlement lag; BR installments (merchant-funded "sem juros" vs customer-funded); `fact_payment_schedule` at one row per settlement tranche.
- Procurement & inventory** — `dim_supplier` (Net 30/45/60, lead time), `fact_purchase_orders`, `fact_inventory_movement` ledger → **derived** `fact_inventory_snapshot` ; moving-average COGS; DIO / DSO / DPO / **CCC** and a company cash-flow timeline.

KPI framework

North Star **Net Revenue (USD)** vs monthly plan. Families: Sales & Growth (AOV, gross margin %, discount rate, return rate, YoY total vs comparable base, revenue vs target, PVM); Payments & Cash (effective fee rate, financing cost, revenue booked vs cash received, DSO); Working Capital (DIO/DSO/DPO/CCC, weeks of cover, aged inventory); Marketing (CTR, CPC, new customers, blended & paid CAC, ROAS, MER, funnel); Web (conversion, AOV, cart abandonment); Logistics (on-time %, delivery days, perfect-order rate, carrier scorecard); CRM (repeat rate, cohort retention, churn, CLV, CSAT). `fact_target` holds a monthly plan by market for Net Revenue, Orders, New Customers, Blended CAC, Gross Margin %.

Out of scope: full P&L / GAAP, real ad attribution (last-touch at channel only), payroll, real personal data (all synthetic, seeded), real-time / streaming.

02 Data Architecture

Raw, messy, multi-source data → a clean, documented **Kimball star schema** that Power BI consumes.

Medallion design (raw → staging → curated), driven by a **seeded, reproducible** Python pipeline (single seed in `etl/config.py` → byte-identical re-runs). Each layer has one job; every script fully rewrites its outputs (idempotent, no partial state).

Architecture (Mermaid source):

flowchart LR

```

subgraph SRC["Source systems (simulated)"]
    Storefront/OMS · ERP · Procurement · WMS · PSP · Web analytics ·
    Google Ads · Meta Ads · CRM · Support desk · Carrier tracking · Finance (FX & targets)
end

SRC -->|01_generate_raw.py| RAW["data/raw/ - Bronze"]
RAW -->|02_clean_stage.py| STG["data/staging/ - Silver"]
STG -->|03_build_curated.py| CUR["data/curated/ - Gold (star schema)"]
CUR --> MOD["Power BI semantic model"] --> RPT["Executive report"]
CUR -. ->|dq_checks.py| Q["data/quality/dq_report.md"]

```

- **Bronze** `data/raw/` — analyst-style exports from 12 simulated systems with deliberately injected defects: category spellings ×3, money strings, 5 date formats + Excel serials, a Latin-1 BR file, 6 SKUs missing from the ERP master, a mislabelled PSP fee column, snapshot-vs-ledger drift, negative sessions. *Nothing downstream writes here.*
- **Silver** `data/staging/` — one typed `stg_*` table per entity: parsing & typing, de-duplication, currency normalization, category/brand standardization, ad-platform union, carrier-JSON flatten, customer-identity resolution, PSP↔order reconciliation. Every altered row is logged to the DQ report.
- **Gold** `data/curated/` — 16 dimensions + 12 facts (Parquet **and** CSV). Moving-average COGS from PO receipts; `fact_payment_schedule` explodes installments; `fact_inventory_snapshot` derived from the movement ledger via a running-balance window function; USD conversion on every money column; `is_comparable_base` = US.
- **DQ gate** — all hard checks pass: no null keys, full referential integrity, `net = gross - discount`, lines roll up to headers, payment schedule ties to order totals, comparable base = US, FX sane, volume / AOV / margin within bands. A violation fails the build.

03 Data Dictionary — inventory

Auto-generated from the files on disk (`etl/gen_data_dictionary.py`). This summary lists the **tables**; the full column-level dictionary is in `03_data_dictionary.md` .

Raw (~19 files / 12 systems): `fx_rates.csv` , `suppliers.csv` , `erp_products.xlsx` (products / cost_history / price_history), `crm_customers.csv` , `crm_customers_br.csv` (Latin-1), storefront orders & order-lines, procurement POs, WMS movements & snapshots, PSP transactions, web analytics, Google/Meta ad exports, support tickets, carrier tracking (nested JSON).

Staging: 20 typed `stg_*` tables (one per meaningful entity).

Curated — 16 dimensions: `dim_date` , `dim_market` , `dim_customer` , `dim_product` , `dim_supplier` , `dim_channel` , `dim_campaign` , `dim_payment_method` , `dim_carrier` , `dim_currency` , `dim_warehouse` , `dim_metric` , `dim_order_status` , `dim_return_reason` , `dim_ticket_category` , `dim_device` .

Curated — 12 facts (grain): | Fact | Grain | |---|---| | `fact_order_lines` | order × product × line_type || `fact_orders` | order header || `fact_payment_schedule` | order × settlement tranche || `fact_purchase_orders` | PO × SKU || `fact_inventory_movement` | stock movement || `fact_inventory_snapshot` | month-end × SKU × warehouse || `fact_web_traffic_daily` | date × market × channel × device || `fact_marketing_spend` | date × market × channel × campaign || `fact_returns` | return line || `fact_support_tickets` | ticket || `fact_exchange_rate` | date × currency || `fact_target` | month × market × metric |

Realised scale: 294 SKUs · ~103k customers · ~158k non-cancelled orders · ~262k order lines · ~14k PO lines · 24 months.

04 ETL Pipeline

`etl/` holds the full four-stage pipeline — `config.py`, `utils.py`, `01_generate_raw.py`, `02_clean_stage.py`, `03_build_curated.py`, `dq_checks.py`, `run_pipeline.py`, `gen_data_dictionary.py`. Seeded and reproducible; ~10 minutes end to end.

Stage	Output	Work
1 · Raw	19 files / 12 systems	Emit source-faithful exports with injected mess
2 · Staging	20 typed <code>stg_*</code> tables	Parse/type, dedupe, normalize currency & categories, union ad platforms, flatten carrier JSON, resolve customer identity, reconcile PSP↔orders — every fix counted into a DQ fragment
3 · Curated	16 dims + 12 facts (Parquet + CSV)	Moving-average COGS from PO receipts; explode installments; derive inventory snapshot from the movement ledger; USD on every money column; <code>is_comparable_base</code> = US
4 · DQ gate	<code>data/quality/dq_report.md</code>	Hard assertions (keys, RI, accounting identities, bands); build fails on violation

Engineering techniques on show: running-balance window function per SKU-warehouse for the month-end snapshot; deterministic seeding; multi-format date parsing (incl. Excel serials); Latin-1 decoding; fuzzy customer-identity resolution. Calibrated over 4 regeneration passes.

05 Data Model

Curated layer is a **Kimball-style star**: one row per business event in the facts, conformed dimensions with integer surrogate keys, a real date dimension, every monetary column stored in both `*_local` and `*_usd` (converted at the transaction-date rate).

- **28 data tables** (16 dims + 12 facts), each with a `Parquet.Document` import partition driven by a `pDataFolder` parameter (model-as-code).
- **~65 relationships**; secondary date roles kept **inactive**, activated with `USERRELATIONSHIP` inside specific measures.
- **Disconnected helper tables**: `Reporting Currency` (USD/EUR/GBP/BRL selector), `Funnel Stage`, `Revenue Bridge`, `PL Line` — drive slicers and waterfall visuals.
- Facts are **never joined to each other**; different grains of the same process are sliced by shared dimensions.

The bus matrix, fact-by-fact grain & measure list, and the full Power BI relationship list are in `05_data_model.md`.

06 DAX Measure Library

167 measures in 14 display folders: 00 Selectors , 01 Sales , 02 Time Intelligence , 03 Comparable Base , 04 Targets vs Plan , 05 Reporting Currency , Payments & Cash , Price / Volume / Mix , Marketing , Website & Funnel , Logistics , 03 Working Capital , CRM , Support & Returns . All monetary measures are USD; *_local columns stay on the detail tables.

Key definitions

Measure	Definition
Net Revenue (USD)	Gross Sales – Returns (Gross Sales = Σ net_amount_usd , excl. cancelled)
Gross Profit (USD)	Net Revenue – COGS + Returns COGS Recovered
Gross Margin %	Gross Profit ÷ Net Revenue
Contribution Margin (USD)	Gross Profit – Marketing Spend – Shipping Cost – Total Payment Cost
Contribution Margin % (Trailing 12M)	12-month rolling CM ÷ 12-month rolling Net Revenue
Net Revenue YoY %	(Current – PY) ÷ PY via SAMEPERIODLASTYEAR
Comparable Base / Expansion	restrict to markets live the whole CY+PY span (US); Expansion = Total – Comparable Base
CCC (days)	DIO + DSO – DPO, period-consistent (balance × days-in-period ÷ COGS-of-period)
Blended CAC (USD)	Marketing Spend ÷ New Customers
Exec Insight	dynamic one-paragraph narrative measure that re-writes itself with the date slicer and every filter

Full DAX + format string for every measure is in 06_dax_measures.md .

07 Report Guide

7 pages, ~150 visuals, `pbir validate --all` passes. **Design identity:** deep-blue corporate tone, one accent; **signature** on every page — navy logo bar + accent rule, then the page title with the Currency and Market slicers, an Analysis-Period date slicer, one row of 6 KPI cards, then two content rows on the 3-30-300 detail gradient. Theme `powerbi/theme/VoltEdge.json` (quiet gridlines, no drop shadow on charts). The whole report **re-denominates** into USD / EUR / GBP / BRL from one control, with a dynamic currency symbol.

#	Page	The question it answers
1	Executive Summary	Are we hitting plan, is growth real, where does the money go?
2	Sales Performance	Revenue booked vs cash received; price/volume/mix; orders
3	Marketing & Acquisition	Funnel, spend vs CAC, new customers by channel
4	Website & Digital	Sessions, conversion, AOV, device mix
5	Logistics & Fulfillment	On-time %, delivery days, cash conversion cycle, carriers
6	CRM & Customer	Cohort retention, churn, repeat rate, support
7	Product & Inventory	SKU-level margin, return rate by category, weeks of cover

Executive Summary carries: KPI row with coloured "**vs Target**" deltas (green/red on the same scale as the bridges), the dynamic **Key Insights** paragraph, a monthly + trailing-12-month **contribution-margin** chart, **four bridges** (net revenue by category, net revenue by region, contribution margin by category, contribution margin by region) and the **Budget scorecard**.

08 Data Quality

Issues are handled in **staging**; every fix is written to the generated `data/quality/dq_report.md`. The **hard checks** (`dq_checks.py`) fail the build on violation: no null keys, full referential integrity, `net = gross - discount`, order lines roll up to headers, payment schedule ties to order totals, comparable base = US, FX within sane bands, and volume / AOV / margin inside expected ranges.

Accepted limitations: campaign performance is media-metrics only (no order↔campaign link in the data — last-touch at channel level); a blank `dim_date` member can appear on cash settling past the window end (fix deferred to widening `DIM_DATE_END`); all customers are synthetic (`Faker`, seeded). Full list in `08_data_quality.md`.

09 Project Walkthrough

Condensed build timeline (full blow-by-blow in `walkthrough.md`):

1. **Scope** locked — English deliverable, `.pbip` (PBIR), ~3 yr of data, multi-country, CSV+Parquet.
2. **Scaffold** — folder structure + README.
3. **Business context & KPI framework** (01) — company, markets, model, KPIs, personas.
4. **Data architecture** (02) — medallion design, 12 source systems, injected-defect catalogue, Mermaid diagram.
5. **Scope revision** — window shortened to 24 months for a clean YoY; expansion dates moved *inside* the window; payments and procurement/inventory deepened.
6. **ETL built & run** — four stages, seeded, calibrated over 4 regeneration passes; all hard DQ checks pass.
7. **Data docs** — dictionary (auto), ETL narrative, data model, data-quality doc.
8. **Semantic model** (model-as-code) — TMDL for 28 tables, ~65 relationships, disconnected helpers.
9. **Report pages** — 7 pages built from JSON bundles via the `pbir` CLI.
10. **Measure & report docs** — auto-generated DAX doc + report guide.
11. **Design passes v1 → v10** — committed a deep-blue identity + logo; header image; combo charts; date slicer; real profitability trajectory (marketing efficiency ramp + unit-cost drift); budget on every KPI; **KPI cards redesigned** — value + colour-coded "vs Target" delta on a white rounded frame; **four bridges**; period-consistent CCC; plan rebased to prior-year × growth.

Current figures (from the latest data generation, full window / USD): Net Revenue ≈ **\$31.7M**; YoY total > **+100 %** (expansion-driven), like-for-like ≈ **+40 %** (US); gross margin ≈ **18 %** (last 12 m ≈ +2 pp); contribution margin ≈ **-3 % → +3 %** (prior vs last 12 months); on-time delivery ≈ **94 %**, avg **4.5 days**; new customers **-14 %** vs target; blended CAC ≈ **\$28** (target \$32.56); repeat-purchase rate ≈ **38 %**.

10 Video Walkthrough Script

A ~3:00 portfolio narration (English, AI text-to-speech) over a screen recording of the report — built to show knowledge of the company, of the data & business, a clear storyline, and visual craft. Full script with timings and shot list in `video_walkthrough_script.md`.

Scene	Beat
1 · Context	Four markets, 24 months of data, medallion → star schema → 160+ DAX measures
2 · Headline	Net revenue ≈ \$32M, ~+3 % vs plan; KPI cards show a colour-coded gap to target
3 · Profitability	Headline growth is expansion; like-for-like ≈ +40 %; margin is the real story (CM -3 % → +3 %)
4 · Four bridges	Where the money comes from — net revenue and contribution margin, by category and by region
5 · Watch item	New customers ~14 % below target; CAC is fine — it's a volume problem, not price
6 · Operations	On-time ≈ 94 %, ~4.5 days; the cash conversion cycle broken into DIO / DSO / DPO
7 · Depth	Payment fees, moving-average COGS, cohorts, SKU margin; whole report re-denominates by currency

Numbers are deliberately rounded so small data-refresh drift doesn't break the narration.